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Case Study 1

Data for Questions 1 to 5

Tutoring Session Readiness Score

An online tutoring platform uses a small feedforward network to estimate a session-readiness score before assigning a tutor. The model uses two input features:

x_1 : student preparation score, x_2 : topic difficulty score.

For one session, $x_1 = 2$ and $x_2 = 1$. The network computes hidden quantities first and then combines them:

$$z_1 = 2x_1 + x_2, \quad z_2 = x_1 - x_2 + 1,$$

$$\hat{y} = z_1 + 2z_2.$$

The computation is feedforward: inputs are used to compute z_1, z_2 , and these hidden quantities are then used to compute $\hat{y}$.

1. What is the value of z_1 for this session?

- ☐ 6
- ☒ 5
- ☐ 4
- ☐ 3

2. What is the value of z_2 ?

- ☐ 0
- ☐ 1
- ☐ 3
- ☒ 2

3. What is the final value of $\hat{y}$?

- ☒ 9
- ☐ 7
- ☐ 8
- ☐ 6

4. Which computation must be completed before $\hat{y}$ is evaluated?

- ☐ The loss must be minimized over all stored tutoring sessions.
- ☐ The target value must be changed by the model before scoring.
- ☒ The hidden quantities z_1 and z_2 must be computed.

5. What makes this a feedforward computation?

- ☐ The output is repeatedly fed back as a new input.
- ☒ Information flows from inputs to hidden quantities to output.
- ☐ The target label is used before the input features.
- ☐ The network stores a separate state for each previous session.

Case Study 2

Data for Questions 6 to 10

Inventory Forecast Calculation Review

A store wants to understand how a change in one product-level input can affect a final forecasting loss. Before the graph is

$$u = 2x, \quad v = u + 3, \quad L = v^2.$$

For the current product, $x = 1$. The team computes local derivatives using the chain rule. Here, x is a simplified d

6. What is the value of u ?

- ☒ 2
- ☐ 1
- ☐ 3
- ☐ 4

7. What is the value of v ?

- ☐ 3
- ☐ 4
- ☐ 6
- ☒ 5

8. What is the loss L ?

- ☐ 20
- ☐ 15
- ☒ 25
- ☐ 10

9. What is $\frac{\partial \mathcal{L}}{\partial v}$ at $v = 5$?

- ☐ 5
- ☒ 10
- ☐ 25
- ☐ 2

10. What is $\frac{\partial \mathcal{L}}{\partial w}$?

- ☐ 10
- ☐ 5
- ☐ 40
- ☒ 20

Case Study 3

Data for Questions 11 to 15

Sensor Calibration Error Analysis

A lab calibrates a sensor whose raw reading must be converted into a corrected measurement. To check the bias, they take two readings: $x = 2$, $y = 5$.

The model is

$$h = w_1 x + b, \quad \hat{y} = w_2 h,$$

with $w_1 = 1$, $b = 0$, $w_2 = 2$. The loss is

$$\mathcal{L} = \frac{1}{2} (\hat{y} - y)^2.$$

Backpropagation computes gradients from the loss back through the output and hidden computations. The gradients are:

11. What is the hidden value h ?

- ☐ 1
- ☐ 4
- ☒ 2
- ☐ 5

12. What is the prediction $\hat{y}$?

- ☐ 3
- ☐ 5
- ☐ 2
- ☒ 4

13. What is $\frac{\partial L}{\partial y}$?

- ☒ -1
- ☐ 1
- ☐ -2
- ☐ 2

14. What is $\frac{\partial L}{\partial w_1}$?

- ☐ -1
- ☒ -2
- ☐ 2
- ☐ 1

15. What is $\frac{\partial L}{\partial w_2}$?

- ☐ -1
- ☐ 1
- ☒ -4
- ☐ 4

16. What is the initial prediction $\hat{y}$?

- ☐ 4
- ☒ 3
- ☐ 5
- ☐ 2

17. What is $\frac{\partial \mathcal{L}}{\partial y}$?

- ☒ -3
- ☐ 3
- ☐ -6
- ☐ 6

18. What is $\frac{\partial \mathcal{L}}{\partial w_2}$?

- ☐ 3
- ☐ -3
- ☐ 6
- ☒ -6

18. What is $\frac{\partial L}{\partial w_2}$?

- ☐ 3
- ☐ -3
- ☐ 6
- ☒ -6

19. What is the updated value of w_1 ?

- ☐ 0.7
- ☐ 1.6
- ☒ 1.3
- ☐ 0.4

20. What is the updated value of b ?

- ☒ 0.3
- ☐ -0.3
- ☐ 0.6
- ☐ -0.6

You may submit any number of times before the due date. The final submission will be considered for evaluation.